Astro 331 Equations **2025 Fall**

constants

$c = 299\,792\,458\,\mathrm{m/s} = \lambda f$ $\quad k = 1.381 \times 10^{-23}$ J/K

$R_\oplus = 6378.137\,\mathrm{km}$ $\quad \mu_\oplus = 398\,600.5\,\mathrm{km^3/s^2}$ $\quad M_\oplus = 7.96 \times 10^{15}\,\mathrm{T\,m^3}$

$273.15\,\mathrm{K} = 0\,^\circ\mathrm{C}$ $\quad \sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$

$1\ \mathrm{AU} = 149\,597\,870.7\,\mathrm{km}$ $\quad S_\odot = \dfrac{L_\odot}{4\pi R^2} = 1358\,\mathrm{W/m^2}$ at 1 AU

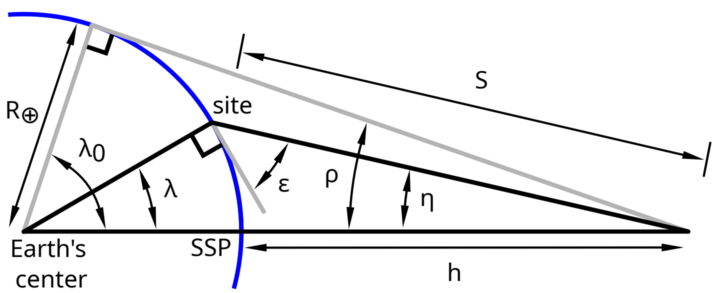
$L_\odot = 3.83 \times 10^{26}$ W (total solar output)

orbits

$$\varepsilon = \frac{V^2}{2} - \frac{\mu}{R} = \frac{-\mu}{2a} \qquad R = \frac{a(1-e^2)}{1 - e\cos\nu} \qquad V_{circ} = \sqrt{\frac{\mu}{R}}$$

$$\mathbb{P} = 2\pi\sqrt{\frac{a^3}{\mu}} \qquad n = \sqrt{\frac{\mu}{a^3}} = \frac{2\pi}{\mathbb{P}} \qquad R = R_\oplus + h$$

mission geometry

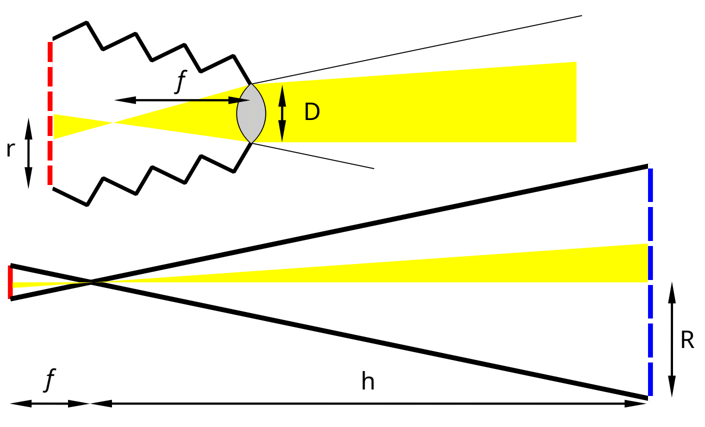


$$\sin\rho = \cos\lambda_0 = \frac{R_\oplus}{R_\oplus + h} \qquad \lambda_0 + \rho = 90^\circ \qquad \eta + \lambda + \varepsilon = 90^\circ$$

$$\tan\eta = \frac{\sin\rho\sin\lambda}{1 - \sin\rho\cos\lambda} \qquad \sin\eta = \cos\varepsilon\sin\rho$$

$$S = R_\oplus\frac{\sin\lambda}{\sin\eta} = -R_\oplus\sin\varepsilon + \sqrt{(R_\oplus\sin\varepsilon)^2 + h^2 + 2R_\oplus h}$$

remote sensing



$$Res = \frac{2.44\lambda h}{D} \qquad Res = \frac{2R}{\#\ pixels} \qquad mag = \frac{f}{h} = \frac{r}{R}$$

$$\lambda_{max} = \frac{2.898 \times 10^{-3}}{T}$$

electrical power

$$T_e = \mathbb{P}\frac{2\rho}{360^\circ} \qquad P_{req} = \frac{P_eT_e/\eta_e + P_dT_d/\eta_d}{T_d} \qquad C_{batt} = \frac{P_eT_e}{DOD\ \eta}$$

$$P_{BOL} = S_\odot A\eta\cos i \qquad P_{sa} = P_{BOL}(1 - degradation)^{time}$$

astrodynamics (angular rates in rad/s, distances in km)

$$\dot\Omega_{J2} \approx -4.171 \times 10^7\frac{\cos i}{a^{7/2}} \qquad \dot\omega_{J2} \approx 2.085 \times 10^7\frac{4 - 5\sin^2 i}{a^{7/2}}$$

$$\dot\Omega_{☾} \approx -7.87 \times 10^{-17}a^{3/2}\cos i \qquad \dot\omega_{☾} \approx 3.72 \times 10^{-17}a^{3/2}(4 - 5\sin^2 i)$$

$$\dot\Omega_\odot \approx -3.58 \times 10^{-17}a^{3/2}\cos i \qquad \dot\omega_\odot \approx 1.79 \times 10^{-17}a^{3/2}(4 - 5\sin^2 i)$$

communication

$$X\ (dB) = 10\log_{10} x\ (linear) \qquad 10^{X\ (dB)/10} = x\ (linear)$$

$$\frac{E_b}{N_0} = \frac{P_tG_tL_lL_sL_mG_r}{kT_rR} = SNR\frac{B}{R} \qquad EIRP = P_tG_tL_l$$

$$\frac{E_b}{N_0} = P_t + G_t + L_l + L_s + L_m + G_r + 228.6 - 10\log_{10}T_r - 10\log_{10}R$$

$$L_s = \left(\frac{\lambda}{4\pi S}\right)^2 \qquad L_m = L_{pt,\,tx} + L_{pt,\,rx} + L_{pol} + L_{atm} + L_{rain}$$

$$L_{pt}\ (dB) = -12\left(\frac{\theta}{\theta_{3dB}}\right)^2 \qquad L_{pol} = \cos^2\theta \text{ (linear pol only)}$$

$$L_{atm}\ (dB) \approx \frac{A_{zenith}(dB)}{\sin\varepsilon} \qquad \theta_{3dB,\,par}\ (\deg) = 70\frac{\lambda}{D_{par}}$$

$$G_{par} = \left(\frac{\pi D_{par}}{\lambda}\right)^2\eta \qquad \eta_{par} = \ 0.55\text{–}0.7 \text{ (typical)}$$

attitude

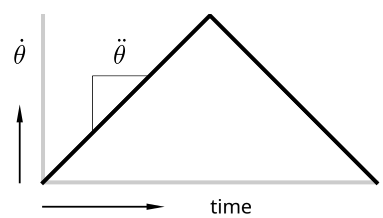
$$\vec{T} = I\vec{\alpha} \qquad \int\vec{T}dt = \Delta\vec{H} \qquad \sum T_{ext} = T_{SRP} + T_{drag} + T_m + T_g$$

$$T_{SRP} = F_{SRP}(c_{ps} - cg) \qquad F_{SRP} = \frac{S_\oplus}{c}A_s(1+q)\cos i$$

$$T_{drag} = F_{drag}(c_{pa} - cg) \qquad F_{drag} = \frac{1}{2}\rho C_dAV^2$$

$$\vec{T}_m = \vec{D} \times \vec{B} \qquad D = NIA \qquad B_{pole} \approx \frac{2M}{R^3} \qquad B_{equator} \approx \frac{M}{R^3}$$

$$T_g = \frac{3\mu}{2R^3}|I_{yaw} - I_{other}|\sin 2\theta$$



reaction wheels

$$\vec{H} = I\vec{\omega} \qquad I_{cylinder,\,z} = \frac{1}{2}mr^2 \qquad T_{slew} = \frac{4I\theta}{t^2}$$

thermal

$$Q_{out} + P_{out} = Q_{in} + Q_{internal} \qquad Q = \varepsilon\sigma AT^4 \qquad Q_\odot = S_\odot A\alpha\cos i$$

$$Q_{albedo} = S_{albedo}A\alpha\cos i\left(\frac{R_\oplus}{R_\oplus + h}\right)^2 \qquad S_{albedo} = 407\ \mathrm{W/m^2}$$

$$Q_{\oplus\ IR} = S_\oplus A\varepsilon\cos i\left(\frac{R_\oplus}{R_\oplus + h}\right)^2 \qquad S_{\oplus\ IR} = 237\ \mathrm{W/m^2}$$

reliability

$R = \mathrm{e}^{-\lambda t} \qquad \lambda = \frac{1}{MTBF} \qquad F = 1 - R$

series : $R = R_A R_B$ parallel : $R = 1 - [(1 - R_A)(1 - R_B)]$